

Akshaj Nevgi

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EDUCATION

Northeastern University, Boston, Massachusetts

Sep 2024 – Dec 2025

Master of Science in Data Science

GPA: 3.96/4.0

Relevant Coursework: Machine Learning, Natural Language Processing, Artificial Intelligence, Data Mining, Machine Learning Operations (MLOps), Database Management Systems

Vellore Institute of Technology, Vellore, India

Sep 2020 – Jun 2024

Bachelor of Technology in Computer Science and Engineering

GPA: 8.79/10.0

Relevant Coursework: Object Oriented Programming in Java, Algorithms, Probability and Statistics, Data Visualization

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern

Feb 2024 – May 2024

COMnet Solutions

Mumbai, India

- Delivered an **LLM-powered** financial knowledge assistant using **Llama 2** and **RAG**, reducing analyst research time by 40%.
- Evaluated transformer-based embedding models on vector databases, optimizing **semantic search** over millions of financial documents.
- Deployed ML models as scalable APIs on **AWS SageMaker**, serving **800+ analysts** with efficient resource use and low latency.
- Built **CI/CD** pipelines with **Jenkins** and automated evaluation, reducing release errors by 20% and ensuring compliant LLM deployments.

Research Assistant

Jul 2023 – Dec 2023

Vellore Institute of Technology

Vellore, India

- Developed a **deep learning pipeline** for retail product recognition by integrating **YOLOv5** with object detection refiner leveraging sub-graph isomorphism, achieving **89% F1-score**.
- Implemented an **OCR-based** Convolutional Neural Network (**CNN**) with dense layers and sequence-to-sequence decoding for text recognition, delivering an average **18% precision improvement** over baseline methods [[Publication Link](#)].

Software Engineer Intern – AI/ML

Jan 2023 – May 2023

Exegesis Infotech

Navi Mumbai, India

- Processed **500K+ monthly** insurance policy transactions from **PostgreSQL** using **AWS Glue** and **PySpark** for downstream analytics.
- Built **RESTful APIs** using **Flask** to serve lead scoring predictions to sales teams, reducing manual prospect qualification time by 30%.
- Performed customer segmentation on **20K+ active policyholders** using KMeans clustering to identify 5 distinct behavioral personas.
- Developed pytest test suite with **90% code coverage** across data pipelines and API endpoints, reducing production bugs by 25%.

Data Scientist Intern

Sept 2022 – Dec 2022

Rydot InfoTech

Gujarat, India

- Applied ML classification models (logistic regression, decision trees, XGBoost) using Python and Scikit-learn that **improved fraud detection accuracy by 27%**, securing client insurance assets.
- Optimized **ETL processing** and data ingestion by 35% in **GCP** and **Snowflake**, accelerating scalable ML deployment and reducing latency.
- Designed **Power BI** dashboards to monitor claims, customer behavior and risks, delivering actionable predictive insights to executives.

PROJECTS

Intelligent Onboarding Assistant | Python, LangChain, Airflow, Docker | [GitHub Link](#)

Sep 2025 – Dec 2025

- Created a production-ready Retrieval-Augmented Generation (**RAG**) system with hybrid retrieval and cross-encoder re-ranking, processing 1000+ pages and 50+ video transcripts to optimize onboarding query response time.
- Built an **MLOps pipeline** using Apache **Airflow** and **FastAPI**, automating data ingestion and deployment on **Google Cloud Platform**.

Healthcare Data Query System | Python, FastAPI, Snowflake, React | [GitHub Link](#)

Jan 2025 – May 2025

- Orchestrated a full-stack healthcare analytics assistant enabling natural language querying using Gemini 2.0 Flash and LangChain agents.
- Processed payer provider cost data in **Snowflake** and created a Pinecone vector database (MiniLM embeddings) for semantic retrieval.
- Engineered multi-agent pipelines to interpret user queries, generate and execute **SQL** and return insights with automated visualizations.

Enhanced Text Detection in Natural Scenes | Python, OpenCV, Flask | [Publication Link](#)

Apr 2023 – Dec 2023

- Developed a Flask-based service integrating MSER feature detection, Sobel edge detection and Norm-CLAHE image contrast enhancement to achieve **93.47% accuracy** in text localization and recognition across 1,500+ images with 11,400+ text instances.
- Crafted algorithms for merging overlapping text regions and filtering non-text components, achieving **15% reduction in false positives**.

TECHNICAL SKILLS

Languages & Databases: Python, SQL, C/C++, Java, JavaScript, HTML, CSS, PostgreSQL, MySQL, MongoDB

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch, Keras, LangGraph, OpenCV, Streamlit, Plotly

Cloud & Platforms: AWS (S3, EC2, Lambda, Glue, SageMaker), Snowflake, Databricks, Kubernetes, GCP, Git, Azure

AI/ML: LLMs, Generative AI, RAG, Prompt Engineering, Regression, Classification, Clustering, Decision Trees, Ensemble Methods

Certifications: AWS Certified Cloud Practitioner, Modern Application Development (Java Spring Boot)